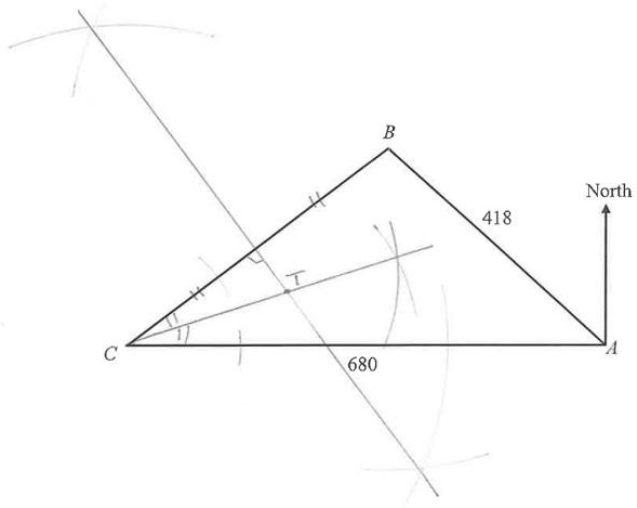
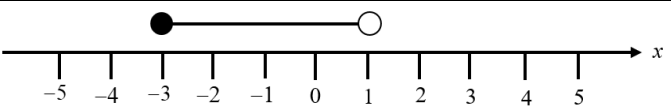
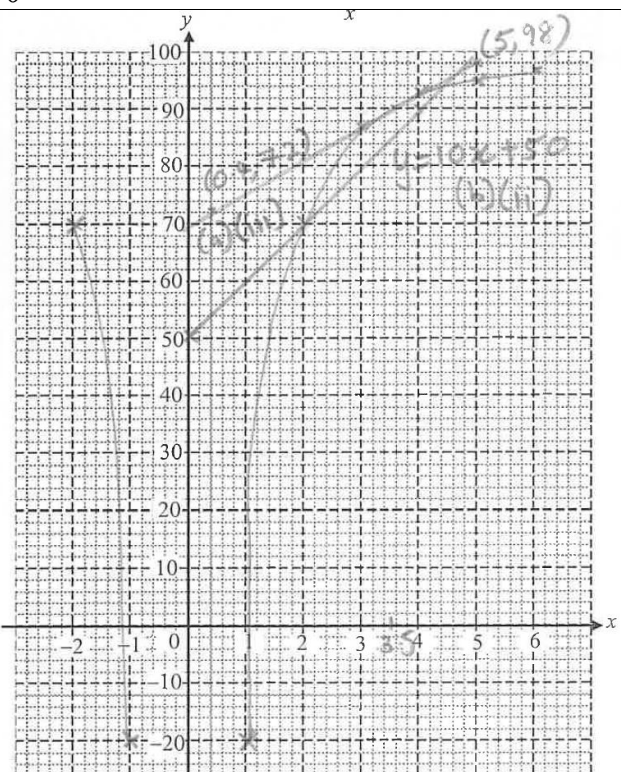


Ngee Ann Secondary School
Secondary 4&5 Mathematics-O
2026 Prelim Examinations Paper 2
Marking Scheme

1(a)	$\angle BAC = 312^\circ - 270^\circ = 42^\circ$ $BC^2 = 418^2 + 680^2 - 2(418)(680)\cos 42^\circ$ $BC = \sqrt{214661.0296}$ $= 463.315$ $= 463.3 \text{ m (to 1 d.p.)}$	[M1] [M1: at least 2 d.p.] [AG]
1(b)	Let the drone be at point S . Let the point above C and at eye level of surveyor be point E . $\tan 16.5^\circ = \frac{SE}{680}$ $SE = 680 \tan 16.5^\circ$ $= 201.425\dots$ Height of drone above C $= 201.425 + 1.74$ $= 203.165$ $= 203.2 \text{ m (1 d.p.)}$	[M1] [A1]
1(c)		[B1: construction of perpendicular bisector] [B1: construction of angle bisector] [B1: correct identification of point T]
1(d)	$\frac{\sin \angle BCA}{418} = \frac{\sin 42^\circ}{463.315}$ $\sin \angle BCA = \frac{418 \sin 42^\circ}{463.315}$ $\angle BCA = \sin^{-1} \left(\frac{418 \sin 42^\circ}{463.315} \right)$ $= 37.1343^\circ$ Bearing of T from C	[M1] [M1]

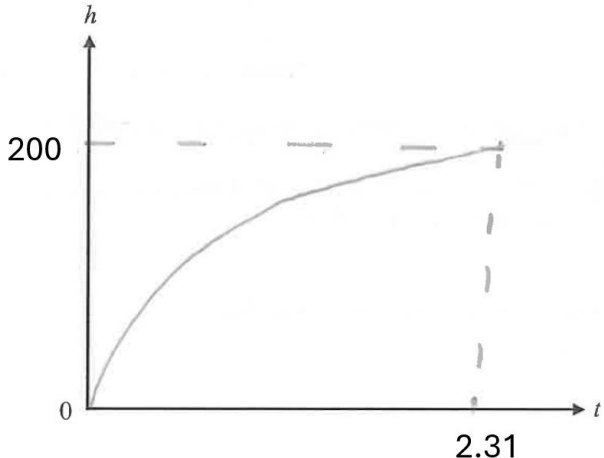
	$= 90^\circ - \frac{37.1343^\circ}{2}$ $= 71.43284^\circ$ $= 71.4^\circ$ Let the foot of perpendicular from T to BC be D. $CD = \frac{463.315}{2}$ $= 231.6575$ $\angle DCT = \frac{37.1343^\circ}{2}$ $= 18.56715^\circ$ $\cos 18.56715^\circ = \frac{231.6575}{CT}$ $CT = 244.377$ $= 244 \text{ m (3 s.f.)}$	[A1] [M1] [A1]

2(a)(i)	$-8 < -5x - 3 \leq 12$ $-5 < -5x \leq 15$ $-3 \leq x < 1$	[M1] [A1]
2(a)(ii)		[B1]
2(b)	$\frac{25}{p} \times \frac{1}{5q^4} \times \frac{pq^4}{5}$ $= 1$	[M1: for $\frac{25}{p}$ or $\frac{1}{5q^4}$] [A1]
2(c)	$\frac{7^{r+1}[7-6]}{7^r}$ or $\frac{7^r[7^2-6(7)]}{7^r}$ $= 7$	[M1] [A1]
2(d)	$\frac{3}{2a-b} - \frac{5a+8b}{4a^2+4ab-3b^2}$ $= \frac{3}{2a-b} - \frac{5a+8b}{(2a-b)(2a+3b)}$ $= \frac{3(2a+3b) - (5a+8b)}{(2a-b)(2a+3b)}$ $= \frac{6a+9b-5a-8b}{(2a-b)(2a+3b)}$ $= \frac{a+b}{(2a-b)(2a+3b)}$	[M1: factorise denominator] [M1: make denominators the same] [A1]

3(a)(i)	70	[B1]
3(a)(ii)		[B1: 7 points plotted correctly] [B1: All points plotted correctly] [B1: Smooth curve passing through all points]
3(a)(iii)	Estimated gradient $= \frac{98 - 72}{5 - 4}$ $= 26$	[M1: drawing of tangent] [A1: accept 5.2 to 6.0]
3(b)(i)	At/After <u>3.5 h</u> of charging, Power Bank A's rate of charging is <u>5.65% per hour</u> .	[B1: allow for ecf]
3(b)(ii)	$x^3 - 5x^2 + 12 = 0$ $x - 5 + \frac{12}{x^2} = 0$ $10x - 50 + \frac{120}{x^2} = 0$ $10x + 50 = 100 - \frac{120}{x^2}$ Draw $y = 10x + 50$ Since the <u>battery percentage after 4h is higher with Power Bank A</u> ($92.5\% > 90\%$), <u>Carl should charge his phone using Power Bank A.</u>	[M1] [M1: drawing of line on diagram] [M1: or any reasonable explanation] [A1]

4(a)	$\overrightarrow{QR} = -\frac{3}{2} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ $= -\frac{3}{2} \overrightarrow{PQ}$ Since $\overrightarrow{QR} = -\frac{3}{2} \overrightarrow{PQ}$, $QR \parallel PQ$, and they <u>have a common point Q</u>, the points P, Q and R are collinear.	[B1] [B1: with conclusion]
4(b)	$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ $\overrightarrow{OP} = \overrightarrow{OQ} - \overrightarrow{PQ}$ $= \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ Since the <u>y-coordinate of P is 0</u>, P lies on the x-axis.	[M1] [M1: with conclusion]
4(c)	$\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR}$ $= \begin{pmatrix} -3 \\ 1 \end{pmatrix} - \frac{3}{2} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ $= -\frac{1}{2} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ $= -\frac{1}{2} \overrightarrow{PQ}$ So, $\overrightarrow{PR} = \frac{1}{2} \overrightarrow{PQ}$, that means $\frac{ \overrightarrow{PR} }{ \overrightarrow{PQ} } = \frac{1}{2}$ $\overrightarrow{PR} : \overrightarrow{PQ} = 1 : 2$ $n = 2$	[M1: accept other reasonable methods, e.g. use of diagram] [A1]
4(d)	$\frac{\text{Area of } \triangle PQS}{\text{Area of } \triangle PRS} = \frac{PQ}{PR}$ $\frac{\text{Area of } \triangle PQS}{10} = \frac{2}{1}$ Area of $\triangle PQS = 20 \text{ units}^2$	[M1] [A1]

5(a)(i)	The <u>title</u> is misleading because it interprets the data for the reader and is <u>biased</u> . In fact, less than 50% of the households (48%) generated less than 400g of food waste that day.	[B1: accept other reasonable answers]
5(a)(ii)	480 g	[B1]
5(a)(iii)	234 g (3 s.f.)	[B1]
5(a)(iv)	In calculating the new estimated mean, the amount of food waste contributed by the additional household, 420g, will be estimated to be 500 g > 480 g . The new estimated mean will be higher than the original one, thus the club is incorrect .	[B1: accept other reasonable answers]
5(b)(i)	$\frac{22}{39} \times \frac{21}{38}$ $= \frac{77}{247}$	[M1] [A1]
5(b)(ii)	$3 \left(\frac{12}{29} \times \frac{17}{28} \times \frac{16}{27} \right)$ $= \frac{272}{609}$	[M1: for the working in bracket] [A1]

6(a)	$4.8 \text{ m}^3 = 4800000 \text{ cm}^3$ Time taken to fill tank without leak $= \frac{4800000}{x} \text{ s}$ Time taken to fill tank with leak $= \frac{4800000}{x-1500} \text{ s}$ $\frac{4800000}{x-1500} - \frac{4800000}{x} = 6000$ $800x - 800(x-1500) = x(x-1500)$ $1200000 = x^2 - 1500x$ $x^2 - 1500x - 1200000 = 0$ (shown)	[M1] [M1: finding difference in time, i.e. LHS of equation] [M1: removing denominator] [A1] [AG]
6(b)	$x^2 - 1500x - 1200000 = 0$ $x = \frac{-(-1500) \pm \sqrt{(-1500)^2 - 4(1)(-1200000)}}{2}$ $x = 2077.59 \text{ or } -577.59$ (2 d.p.)	[M1] [A1, A1]
6(c)	Time taken to fill tank that day $= \frac{4800000}{2077.59 - 1500}$ $= 8310.39 \text{ s}$ $= 2 \text{ h } 19 \text{ min}$ (nearest minute)	[M1] [A1]
6(d)		[B1: correct shape of curve; labeling of axes not necessary]

7(a)(i)	Cost without set meal = $4.80 + 2.80 + 2.00$ = \$9.60 Percentage savings $= \frac{9.60 - 8.50}{9.60} \times 100\%$ = 11.5% (3 s.f.)	[M1] [A1]
7(a)(ii)	$\left(1 - \frac{x}{100}\right)(1.1)(1.09)(8.5) = 9.7$ $1 - \frac{x}{100} = \frac{9.7}{10.1915}$ $\frac{x}{100} = 1 - \frac{9.7}{10.1915}$ $x = 4.8226$ $x = 5$ (nearest whole number)	[M1: correctly accounting for service charge and GST] [M1: correctly accounting for discount] [A1]
7(b)(i)	The elements of P suggest that the cost of main dish, soup and drinks have <u>increased by 20%, 40% and 30% respectively</u> .	[B1: accept other reasonable answer]
7(b)(ii)	$(1.2 \quad 1.4 \quad 1.3) \begin{pmatrix} 2.20 & 1.80 \\ 0.85 & 1.10 \\ 0.40 & 0.40 \end{pmatrix}$ $= (4.35 \quad 4.22)$	[B1; B1]
7(b)(iii)	Percentage profit $= \frac{8.8 - 4.35}{4.35} \times 100\%$ = 102% (3 s.f.) Since <u>102% > 90%</u>, the owner <u>should not</u> revise the selling price of the Chicken Set Meal.	[M1] [A1] [A1: include justification and conclusion]

8(a)	$\angle ABC = \angle CDE = \frac{\pi}{2}$ rad (right $\angle$ in a semicircle) $AB \parallel DE$ because $\angle ABC$ and $\angle CDE$ are <u>alternate angles</u> .	[B1: reason] [B1: correct pair of parallel lines and reason]
8(b)	$\angle CDR = \angle PBC = 0.37$ rad (alt. $\angle$ s) $A: \angle ABP = \angle EDR = \left(\frac{\pi}{2} - 0.37\right)$ rad $\angle BPC = \angle CRD = 2.27$ rad (alt. $\angle$ s) $A: \angle APB = \angle ERD = (\pi - 2.27)$ rad (adj. $\angle$ s on a st. line) By AA, triangle ABP is similar to triangle EDR .	[B1: any correct value for pair of angles, i.e. $\angle APB$ and $\angle ERD$ or $\angle ABP$ and $\angle EDR$] [B1: any correct reason] [B1]
8(c)(i)	$\angle BAC = \pi - \angle ABP - \angle APB$ ($\angle$ sum of Δ) $= \pi - \left(\frac{\pi}{2} - 0.37\right) - (\pi - 2.27)$ $= 1.06920$ $\angle BOC = 2 \times 1.06920 \dots$ ($\angle$ at centre = 2 $\angle$ at circumference) $= 2.1384 \dots$ rad $= 2.138$ rad (3 d.p.)	[M1: reason not needed] [M1: reason not needed] [Al: at least 4 d.p.] [AG]
8(c)(ii)	Arc length $= (8)(2.138 \dots)$ $= 17.1$ cm (3 s.f.)	[B1]

9(a)	Area of hall = $(9 \times 4) \times (6 \times 4)$ $= 864 \text{ m}^2$ Area of stage = $(1.5 \times 4) \times (3 \times 4)$ $= 72 \text{ m}^2$ Area of sound booth = $(0.7 \times 4) \times (2.1 \times 4)$ $= 23.52 \text{ m}^2$ Area of clear zones = $\pi(1)^2 \times 3.5$ $= 10.99557 \text{ m}^2$ Total usable area = $864 - 72 - 23.52 - 10.99557$ $= 757.48443 \text{ m}^2$ Maximum number of standing tickets = 757.48443×1.4 $= 1060.478$ $= 1060$ (nearest whole number)	[M1: area of clear zones] [M1: subtraction] [M1] [A1]
9(b)	Option 1: Revenue = 1060×18 $= \\$19080$ Cost = $1500 + 1072 \times 1.5$ $= \\$3108$ Profit = $19080 - 3108$ $= \\$15972$ Option 2: Usable area excluding safety zone $= 757.48443 - 1.2 \times (6 \times 4)$ $= 728.68443 \text{ m}^2$ Usable area for Seated Zone = $\frac{3}{5} \times 728.68443$ $= 437.21 \text{ m}^2$ Maximum number of seated tickets = 437.21×1 $= 437$ (nearest whole number) Usable area for Standing Zone = $\frac{2}{5} \times 728.68443$ $= 291.47 \text{ m}^2$ Maximum number of standing tickets = 291.47×1.4 $= 408.058$ $= 408$ (rounded down for safety) Revenue = $437 \times 32 + 408 \times 15$ $= \\$20104$ Cost = $1.2 \times 1500 + (442 + 412) \times 1.5$	[M1: correct finding of revenue/cost] [M1] [M1] [M1: correct usable area for Seated or Standing Zone] [M1: correct maximum number of seated or standing tickets]

= \$3081 Profit = $20104 - 3081$ = \$17023 Since $\\$17023 > \\15972, Nate should select Option 2 to maximise profit and raise more funds for charity.	[M1: correct finding of revenue/cost] [A1: correct justification and conclusion]
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